Genetic programming for drone swarm behaviour optimisation

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***Abstract*—This study presents a method of leveraging genetic programming principles to determine the flight parameters of a drone swarm controlled by a bio-inspired algorithm developed by Matthieu Verdoucq. The study explores how this method can be used to optimise the behaviour of a drone swarm in a search-and-rescue scenario, heavily relying on parallelized GPU simulations via PyTorch.**

***Index Terms*—Collective motion, Genetic programming, 3D Flocking Algorithm, Distributed Control, Drone swarm, Unmanned Aerial Vehicle (UAV)**

I. Introduction

Single unmanned aerial vehicles (UAVs) have proven to be exceptionally effective at a wide range of tasks, ranging from cinematography to pesticide application. Swarms of UAVs, although not yet commonly used, hold great promise for future applications. One of these is search and rescue (SAR). In such scenarios a drone swarm has an evident numerical advantage over a single UAV. Controlling the swarm and ensuring it efficiently covers the search area is however a significant challenge.

Our study is based on the bio-inspired flocking algorithm developed by Matthieu Verdoucq for his Ph.D. thesis [1]. A swarm controlled by this algorithm can exhibit emergent behaviours such as schooling (the drones flock together in an orderly fashion), swarming (the swarm remains tight-knit but the drones move erratically), or milling (the drones rotate around swarm's centre in an orderly fashion). Matthieu Verdoucq and his colleagues [2] have demonstrated these behaviours emerge from a specific set of parameters which define the equations used to control the drones.

Our goal is to implement genetic programming paradigms to optimise this set of parameters to adapt the swarm's behaviour to a SAR scenario.

This scenario is modeled by dividing the area to be explored into a grid. The gathering of information is simulated by assigning a numerical freshness value to each cell, which decreases over time.

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II. Related Work

The work presented in this article is a continuation of the research done by Matthieu Verdoucq for his thesis, which he successfully defended in February 2024.

A. Flocking algorithm

The flocking algorithm developed by Matthieu Verdoucq [1] is inspired by the behaviour of a school of Rummy-nose tetra (*H. Rhodostomus*). Tamás Vicsek and Anna Zafeiris [3] were able to extract a model of social interactions between the fish which leads to emergent behaviour. Matthieu Verdoucq then adapted this model to the dynamics of drones by adding, amongst other things, a vertical component to the social interactions.

Lei et al. [4] have shown that Rummy-nose tetra only interact with a limited number of their neighbours. Matthieu Verdoucq implemented this in his interaction model which means the drones need only gather a limited amount of information from their social environment to display coordinated behaviour.

The emergent behaviour of the swarm is influenced by a set of parameters which determine the guidance of each individual drone in the swarm. Verdoucq et al. [5] have shown how changing these parameters can alter the behaviour of the swarm in real-time.

These studies have observed that certain combinations of parameters induce the same behaviour in the swarm. Each of these specific behaviours is called a phase. The research produced by Matthieu Verdoucq for his thesis principally focuses on phases emerging from interactions between the alignment and attraction coefficients γ_{ali} and γ_{att}. Matthieu Verdoucq [1] has identified three distinct phases linked to these parameters: schooling, swarming and milling.

This means that although the paths of individual UAVs in the swarm cannot be controlled, their global behaviour can. However, there are many other parameters which influence the behaviour of the swarm which were not studied by Matthieu Verdoucq. The complexity of the interactions between these parameters prevents a manual tuning of this set of parameters. Genetic programming paradigms have been used to optimise drone swarm objectives and are promising means of accomplishing our objective: optimising the set of drone guidance parameters to a specific task.

B. Genetic programming

Genetic programming algorithms are powerful because they do not require knowing the structure of the solution to find it [6]. The problem at hand: determining the right parameters to induce the desired behaviour in the drone swarm, is therefore very well suited to genetic programming because the interactions between the parameters and their resulting influence on the swarm's behaviour are opaque.

While some studies, such as the work by Sefa Abraç and Tarik Veli Mumcu [7], focus on combinatorial mutation strategies like displacement or scramble mutations for task assignment, optimizing flight dynamics requires a continuous parameter space approach. Our study will therefore adjust the real-valued parameters controlling the drone swarm's behaviour by employing continuous real-parameter optimization techniques. Specifically, we will utilize uniform crossover operations combined with Gaussian noise mutations to ensure both the exploitation of successful traits and the exploration of new behavioural paradigms.

C. Information gathering strategies

A genetic algorithm can only be as good as its fitness function. In the cas of a SAR scenario the fitness depends a lot on how the gathered information is shared between the drones in the swarm.

One such method is pheromone-based stygmergic coordination [8], [9]. This method relies on digital pheromones being released by the drones to mark the cells they have explored. These pheromones decay and disappear over time, representing the information losing relevance as time goes on. The pheromones repel other drones, thus e:w ncouraging the swarm to fly to cells which have yet to be explored. Xuan et al. [9] have even explored how the use of two distinct pheromones in an under water exploration scenario: a short term pheromone to encode the recent activity of the swarm and a long term pheromone to encode the global exploration coverage. They have showed this method can achieve a very high coverage uniformity of the water column.

III. Emergence of Collective Behaviors: Milling and Flocking

A. Defining the Cost Function

To actively search for specific topological phases, the fitness function evaluated by the Genetic Algorithm must explicitly reward desired macroscopic states while penalizing collisions and excessive energy consumption. We define a multi-objective cost function balancing group polarization, milling momentum, and control effort.

- Let N represent the number of drones in the simulation.
- Let $p_i = (x_i, y_i)$ represent the position of drone number i.
- Let ϕ_i represent the heading of drone number i.
- Let v_i represent the speed of drone number i.

The cost of the effort cost c_{effort} is calculated as follows:

$$c_{\text{effort}} = \sum_{i=1}^{N} |\dot{\phi}_i| \tag{1}$$

The center of the swarm B is determined:

$$B = \frac{1}{N} \sum_{i=1}^{N} p_i \tag{2}$$

The distance of each drone to this center is calculated as such:

$$d_i = \|p_i - B\|_2 \tag{3}$$

The cost of the dispersion cost c_{disp} is calculated as follows:

$$c_{\text{disp}} = \left| \left(\frac{1}{N} \sum_{i=1}^{N} d_i \right) - 5.0 \right| \tag{4}$$

The polarisation parameter is determined as follows:

$$P = \sqrt{\left(\frac{1}{N} \sum_{i=1}^{N} \cos \phi_i \right)^2 + \left(\frac{1}{N} \sum_{i=1}^{N} \sin \phi_i \right)^2} \tag{5}$$

This allows for the calculation of the polarisation cost c_{pol}:

$$c_{\text{pol}} = (1 - P) \tag{6}$$

To calculate the collision cost c_{coll}, we first determine the distance between drones:

$$d_{ij} = \|p_i - p_j\|_2 \tag{7}$$

Thus:

$$c_{\text{coll}} = \frac{1}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \mathbb{1}_{(d_{ij} < d_{\text{coll}})} \tag{8}$$

To compute the milling cost c_{mill}, we first compute the speed vector V_i for each drone:

$$V_i = \begin{pmatrix} v_i \cos \phi_i \\ v_i \sin \phi_i \end{pmatrix} \tag{9}$$

Then we determine the average speed of the swarm:

$$\bar{V} = \frac{1}{N} \sum_{i=1}^{N} V_i \tag{10}$$

Thus follows the centred speed:

$$\Delta v_i = v_i - \bar{V} \tag{11}$$

And the centred position:

$$\Delta p_i = p_i - B \tag{12}$$

These are used to determine the angle of the postion to the swarm's centre:

$$\theta_i = \text{atan2}(\Delta p_{i,y}, \Delta p_{i,x}) \tag{13}$$

And the angle of the speed vector to the swarm's centre:

$$\phi_{v,i} = \text{atan2}(\Delta v_{i,y}, \Delta v_{i,x}) \tag{14}$$

These then allow for the computation of the milling cost:

$$c_{\text{mill}} = \left| \frac{1}{N} \sum_{i=1}^{N} \sin(\theta_i - \phi_{v,i}) \right| \tag{15}$$

The final cost c_{tot} function is the sum of all of these costs, to which the exploration cost can be added if relevant (this cost will be detailed in a further section). Each cost c_x is associated to a weight W_x which is used to tune the relative influences of these parameters in the final cost function:

$$\begin{aligned} c_{\text{tot}} = \ & c_{\text{effort}} \times W_{\text{effort}} \\ & + c_{\text{disp}} \times W_{\text{disp}} \\ & + c_{\text{pol}} \times W_{\text{pol}} \\ & + c_{\text{coll}} \times W_{\text{coll}} \\ & + c_{\text{mill}} \times W_{\text{mill}} \\ & + c_{\text{explo}} \times W_{\text{explo}} \end{aligned} \tag{16}$$

B. Parameter Sweep and Scalability

We employ a Genetic Algorithm to discover parameter bounds that reliably trigger the milling phase. However, a set of parameters optimized for $N = 10$ drones may degrade when applied to $N = 30$. Through systematic parameter sweeping, we analyze the scalability of the discovered behavioral regions.

Fig. 1. Correlation matrix of the swarm parameters generated during the sweeping process, highlighting parameter sensitivities for the milling phase.

IV. Application to Exploration and SAR Scenarios

A. The Grid Coverage Problem

In Search and Rescue (SAR) operations, the primary objective translates mathematically to an area coverage optimization problem. The operational space is discretized into a 2D freshness grid. As drones traverse the environment, they update the freshness of the underlying cells. The swarm's performance is intrinsically linked to its ability to maximize the global freshness score over a fixed time horizon.

Fig. 2. Heatmap demonstrating the spatial coverage and freshness distribution of the drone swarm operating under GA-optimized parameters.

B. Fitness Evaluation for Exploration

The exploration fitness relies on two main factors: the continuous discovery of unseen areas and the periodic revisiting of previously explored zones. This mimics the decay of information relevance in dynamic SAR environments.

V. Distributed Information Gathering Strategies

A genetic algorithm can only be as good as its fitness function. In the case of a SAR scenario the fitness depends a lot on how the gathered information is shared between the drones in the swarm.

A. Pheromone-based Stigmergic Coordination

One such method is pheromone-based stigmergic coordination [8], [9]. This method relies on digital pheromones being released by the drones to mark the cells they have explored. These pheromones decay and disappear over time, representing the information losing relevance as time goes on. The pheromones repel other drones, thus encouraging the swarm to fly to cells which have yet to be explored.

Xuan et al. [9] have even explored how the use of two distinct pheromones in an exploration scenario: a short term pheromone to encode the recent activity of the swarm and a long term pheromone to encode the global exploration coverage. They have showed this method can achieve a very high coverage uniformity of the water column.

In our simulation model, this stigmergic coordination is mathematically inverted and implemented through a discrete spatial "spoilage grid". Instead of maximizing freshness, the grid tracks spoilage, starting at a maximum value S_{max} representing completely unexplored territory. As agents traverse the environment, they clear this spoilage. To account for altitude and sensor mechanics, the update is not a simple cell overwrite but a 2D Gaussian splat governed by the Inverse Square Law. The intensity and spread of the clearing effect dynamically depend on the drone's altitude (Z) and its Field of View (FOV).

To simulate the decaying relevance of information over time, the grid undergoes a uniform time-step spoilage accumulation governed by a multiplicative factor μ_{spoil} and an additive factor α_{spoil}:

$$S_{t+1}(x, y) = \min(S_{max}, S_t(x, y) \cdot \mu_{spoil} + \alpha_{spoil})$$

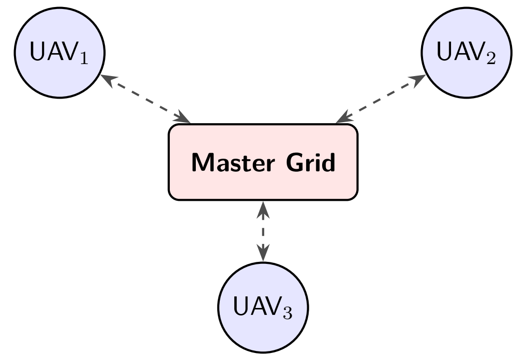
This increasing spoilage acts as an attractive virtual pheromone. Agents compute an exploration gradient vector directing them towards high-spoilage areas, driving the swarm naturally to continuously revisit and explore regions.

B. Impact of Communication Topologies

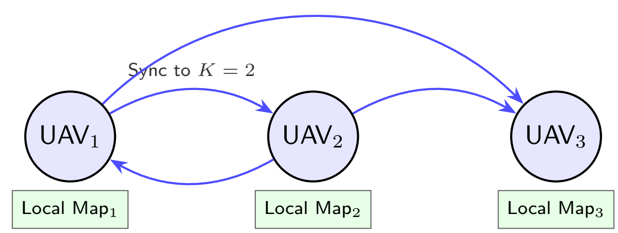
To evaluate the sim-to-real gap regarding network constraints, we compare several information-sharing topologies. The exploration framework relies on two distinct but complementary mechanisms: the map storage and sharing strategy (`MAP_STRATEGY`), and the navigation decision-making (`EXPLO_STRATEGY`).

While a centralized oracle provides theoretical maximum efficiency by granting all agents instantaneous access to a master freshness grid, real-world UAVs rely on distributed mesh networks with limited bandwidth and range. To accurately model these constraints, we implemented and evaluated three distinct communication paradigms for map synchronization:

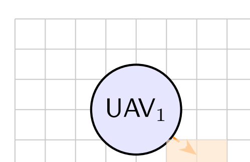
- **Global Oracle (`global`):** Agents have perfect, real-time knowledge of the entire environment's map. Every drone updates and reads from a single, centralized tensor. This serves as our theoretical upper-bound baseline.



- **Local Shared Map (`local_shared`):** Agents maintain individual, localized versions of the map. Synchronization occurs between drones using a topological approach rather than a strictly spatial one. Specifically, map data is shared unilaterally with an agent's K-Nearest Neighbors (e.g., $K = 2$). Furthermore, to simulate bandwidth constraints, this map synchronization is not continuous but occurs at a discrete interval of τ_{sync} ticks (e.g., `REFRESH_MAP_TICKS` $= 50$).



- **Local Individual Map (`local_individual`):** To severely reduce communication overhead and eliminate structural dependencies, drones do not exchange any map data over the network. Instead, each drone relies strictly on its own isolated memory grid, which is updated solely by its own field of view (FOV).



Isolated Local Memory Grid

Regardless of the chosen mapping topology, the actual navigation of the swarm is driven by a local gradient descent approach (`EXPLO_STRATEGY = "local_gradient"`). Each drone computes a spatial gradient vector over its currently available map, indicating the direction of highest spoilage. This vector directly influences the drone's individual yaw command. By evaluating the swarm's coverage efficiency across these three modalities, we can assess the robustness of the genetic algorithm's optimized parameters when subjected to realistic communication bottlenecks.

VI. Discussion and Conclusion

The optimization of bio-inspired swarm parameters through genetic programming presents significant advantages for tailoring complex multi-agent systems to specific operational requirements. Our framework leveraged PyTorch tensor operations to massively parallelize the simulation environments, enabling the rapid evaluation of thousands of swarm configurations simultaneously. By optimizing a multi-objective fitness function, the Genetic Algorithm successfully isolated parameter combinations that balance robust milling behaviour, efficient area coverage, and minimal control effort.

A key contribution of this study is the integration of realistic communication constraints into the evaluation pipeline. The implementation of discrete exploration modes, ranging from a centralized global oracle to decentralized local gradients and delayed map sharing, demonstrates how network limitations directly affect emergent swarm behaviors and global coverage efficiency. Furthermore, our parameter sweeping methodology highlighted the scalability challenges of the swarm, quantitatively showing how parameters optimized for a specific number of agents behave when deployed across varying swarm sizes.

Future work should focus on narrowing the sim-to-real gap. While our current physics engine successfully incorporates inter-agent collision avoidance and basic kinematic updates, transitioning to a fully dynamic quadrotor simulation is the next critical step. This will require introducing sensory noise, more complex communication packet loss models, and high-fidelity aerodynamic constraints. Additionally, implementing dynamic obstacles and individual battery life constraints would further validate the applicability of these bio-inspired control algorithms in real-world Search and Rescue deployments.

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